I am an applied microeconomist, focusing on two overarching themes of research: the impact of weather shocks on socioeconomic outcomes and the influence of public policy changes on socioeconomic and health outcomes. Broadly, my interests lie in labor economics, health economics, environmental economics, development economics, and the economics of crime. In the following passages, I briefly outline my current research and discuss a couple of potential research ideas. For my JMP, I investigate the causal effect of hurricanes on county-level crimes in the US. While, in another working paper, I intend to examine the effects of Bihar's[1] alcohol ban on domestic violence and infant health outcomes.

1 Do Storms Bring Crime? Evidence from US counties

Hurricanes are the most frequently occurring large-scale natural disasters in the US, with the country experiencing an average of 5.6 storms annually between 1980 and 2023. During this period, a total of 248 storms struck the US, resulting in damages worth $1,834.4 billion. This paper investigates the impact of randomly occurring hurricane-level winds on county-level crimes, economic and demographic outcomes, by utilizing a county fixed effects design. I find that, stronger hurricanes produce larger treatment effects across all sets of outcomes. Counties, subjected to major hurricanes witness a 13.3% rise in violent crimes and a 6.5% rise in property crimes. Increases in violent crimes are driven by incidents of robbery and aggravated assault, whereas property crimes tend to increase as a result of vandalism and vehicle theft. The paper finds, that in the immediate aftermath of both major and minor hurricanes, affected counties experience a reduction in total population and unemployment rates, alongside increases in the share of the non-white population, annual wages, number of business establishments, and the poverty rate. However, these initially positive economic effects are reversed in the later part of the decade.

This paper intends to make several key contributions to the current literature that explores the intersection of natural disasters and crime. Firstly, unlike previous research looking at the effect of hurricanes on crimes in the US, (Augusto, 2021; Frailing et al., 2021; Barton et al., 2020; Weil, 2020; Spencer, 2017; Leitner and Helbich, 2011; Schumacher et al., 2010) I do not limit the scope of my study to a particular geographical region or a specific hurricane event. I examine all tropical cyclones originating in the Atlantic basin, between 1990-2018. This larger panel of hurricanes allows me to analyze the short term (1-3 years)[2], medium term (4-9 years), and longer term (10-20 years) effects of hurricanes on crimes in the US. I also utilize a much more granular hurricane exposure dataset, which allows me to vary my key treatment variable by time and intensity across US counties. Secondly, I explore a much broader range of crimes than addressed in earlier studies, allowing for an examination of the heterogeneity in the impact of hurricanes on crimes. Thirdly, I combine rainfall data with hurricane information to differentiate between counties experiencing medium and heavy rainfall, along with hurricane-level winds. Fourthly, by exploring the economic rationale behind hurricane-induced crime, I aim to bridge the gap between two major strands of literature; the labor market effects of natural disasters and the economic causes of crime.

Distinguishing between the crime effects of minor and major hurricanes[3], I find that major hurricanes produce stronger effects on all key outcome variables (Figs. 1 and 2). Major hurricanes

[1] Bihar is the third most populous state in India.

[2] Following a hurricane.

[3] A minor hurricane is assigned a Saffir-Simpson scale of one to two and has maximum sustained surface wind speed between 64-95 knots, while a major hurricane is a category-3 and higher hurricane and has maximum sustained surface wind speed higher than 95 knots.

lead to increases in both property and violent crimes, with property crimes increasing more significantly for coastal counties in the immediate term (1-3 years following a hurricane). In comparison, minor hurricanes produce heterogeneous effects on the major crime indices. Violent crimes fall significantly by 7.5% while property crimes increase marginally by 0.6%, in the immediate aftermath of a minor hurricane.

Beyond examining the impact of hurricanes on crime rates, this study also investigates their effects on various county-level economic and demographic indicators (Fig. 2). This helps to unearth the possible mechanisms, which influence crime rates in the aftermath of a large-scale natural disaster. The findings reveal that, in the immediate aftermath of both major and minor hurricanes, affected counties experience significant reductions in total population and unemployment rates, accompanied by increases in the non-white population share, annual wages, number of firms, and poverty rate. However, in the medium term, the initial positive economic effects dissipate, as wages and the number of establishments decline, and unemployment and poverty rates surge significantly.

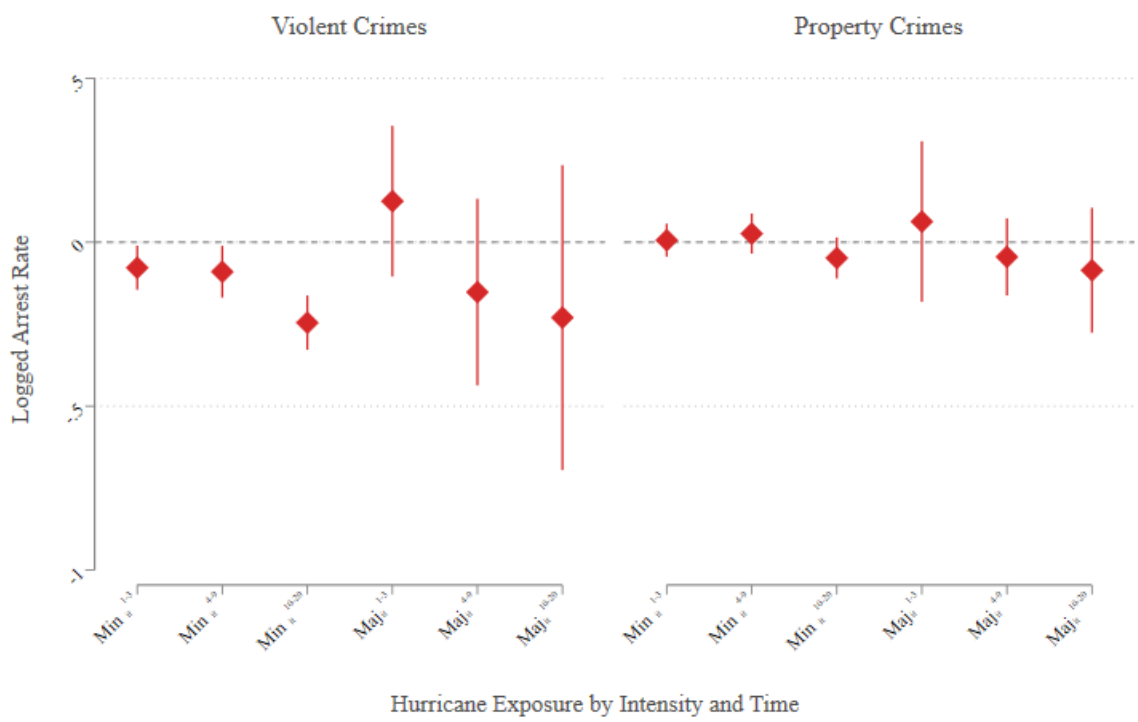

Figure 1: Impact of Hurricane Exposure on Crime-wise Arrest Rates.
Note: Figure plots the estimates and 95% confidence intervals from my baseline fixed effects specification.[4]

[4]For my baseline specification, I construct two key sets of indicator variables, Maj_{it} and Min_{it}. These variables take on values of 0 or 1, depending on whether the county (i) experiences a major or a minor hurricane over the years $t-1$ to $t-20$. Counties experiencing maximum sustained surface winds greater (lesser) than 96 knots, are considered to have experienced a major (minor) hurricane. Parameters β and θ capture the impact of either type of hurricane on the set of outcome variables while controlling for time (μ_t) and county (α_i) fixed effects. Additionally, I control for other factors (X_{it}), which might influence crime rates besides extreme weather events. $Y_{it} = \alpha_i + \mu_t + \beta_1 \text{Min}_{it}^{1-3} + \beta_2 \text{Min}_{it}^{4-9} + \beta_3 \text{Min}_{it}^{10-20} \theta_1 \text{Maj}_{it}^{1-3} + \theta_2 \text{Maj}_{it}^{4-9} + \theta_3 \text{Maj}_{it}^{10-20} + \gamma' \ln(X_{it}) + \epsilon_{it}$.

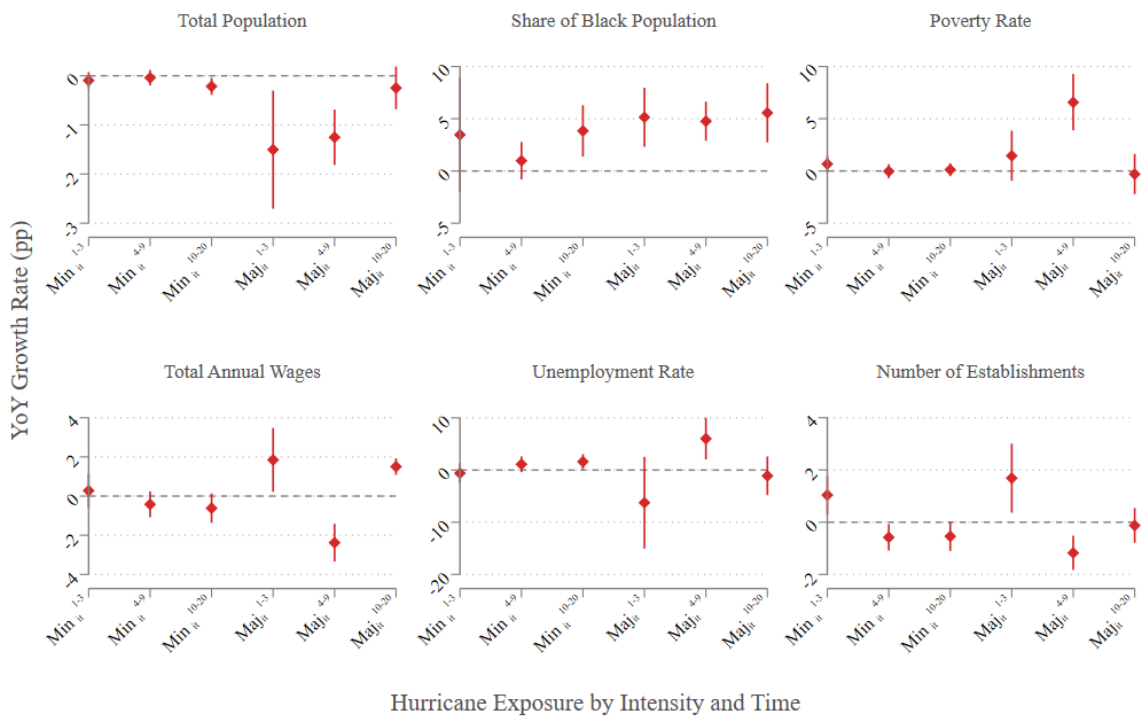

Figure 2: Impact of Hurricanes on Economic and Demographic Outcomes.
Note: Figure plots the estimates and 95% confidence intervals from my baseline fixed effects specification.

This paper takes a modest step towards understanding the interlinkages between extreme weather events, crime, and labor market outcomes. Hence, highlighting the substantial scope for future research in this area. It reveals that, the nexus of natural disasters and crime remains an understudied area of research, particularly from an applied microeconomics perspective. Thus, the next step of this study would be to inform policy-oriented solutions to address the surge in crime following major weather events. Wherein, I would like to utilize the findings of this paper to formulate policies, centered around the deployment of law-enforcement personnel in the aftermath of large-scale natural disasters.

2 Ban the Booze: Effects of Alcohol Ban on Domestic Violence and Infant Health

The Indian state of Bihar issued a ban on all types of alcohol in the state and criminalized the sale and consumption of alcohol within the state, effective from April 5th, 2016. In this paper, I intend to investigate the impact of the ban on domestic violence (DV) and child outcomes. As depicted in Fig. 3, 83% of women who reported a drunken husband also reported intimate partner violence. This statistic underscores the critical importance of studying the relationship between alcohol consumption and domestic violence, and this exogenous policy shock lends us that opportunity. The ban reflected the fulfillment of an earlier election campaign promise, under which the incumbent government had claimed that they will impose a ban on alcohol consumption if voted back to power. This was a prolonged demand of women, chiefly belonging

to economically marginal households, who were most often subjected to domestic violence and had reduced access to criminal justice.

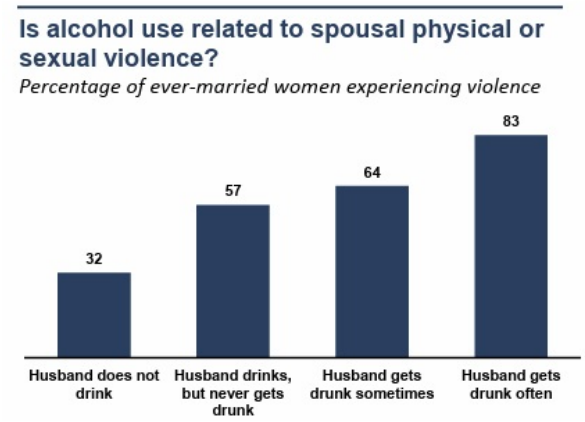

Figure 3: Alcohol Consumption and Domestic Violence in Bihar, India

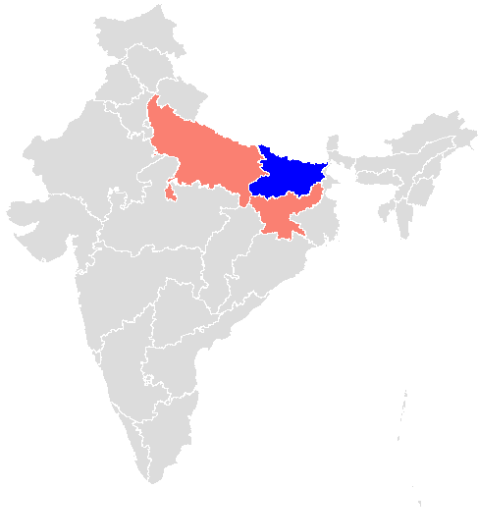

Figure 4: Control and Treatment States

This paper intends to examine three important policy implications. Firstly, does the ban reduce the consumption of alcohol. Secondly, does it reduce incidents of domestic violence.[5] And thirdly, is there an improvement in infants' health outcomes. I plan on utilizing a logistic regression form of econometric specification to answer the first policy question and use a matched Difference-in-Differences (DiD) approach to answer the second and third ones. For the matched DiD approach, I intend to use districts in Jharkhand and western Uttar Pradesh as my control areas. Fig. 4 depicts the control states, colored in salmon, and the treated state, colored in blue. The chief contributions of this paper would be to analyze the efficacy of this policy in terms of its health impact and also to contribute, to prior research which has examined the association between pre-natal exposure to domestic violence and pregnancy outcomes, i.e. low-term birth weight and pre-term birth, Silverman et al. (2006); Janssen et al. (2003); Murphy et al. (2001). By doing so, I will be able to connect my findings to the work of previous researchers who have investigated how poorer infant health endowments affect educational and labor market outcomes in early and later life Currie and Hyson (1999); Case et al. (2005); Johnson and Schoeni (2011)

To achieve the research objectives, I plan to analyze data from two rounds of India's National Family and Health Survey (NFHS): NFHS-4, conducted in 2015 (baseline) before the alcohol ban, and NFHS-5, conducted in 2019 (follow-up), allowing for a comparison of outcomes pre- and post the implementation of the ban. The NFHSs are large household and individual-level surveys. For the 2019 survey round, more than 700,000 women between the ages of 15-49 were interviewed. These surveys implement a separate module for domestic violence, under which women of the households are separately interviewed (without the presence of males of the household) in order to reduce misreporting owing to fear of reprisals. Thus, this provides me with an ideal opportunity to study the effect of the alcohol ban policy on domestic violence. The NFHSs also record detailed health information on everyone belonging to the household, thus allowing me to observe the effect of the policy on both maternal and fetal health.

[5] Domestic violence is one of the biggest consequences of drunkenness in this society, Fig. 3.

3 Working Ideas

3.1 Sugar Shock: Health Impact of Covid-19 Lockdowns

The paper intends to examine the impact of the severity of COVID-19 lockdowns on individual health, as reflected by their blood glucose and blood pressure levels. The study will focus on India, where different states implemented COVID-19 lockdowns of varying intensities from 2020 to 2021. India's NFHS-5 presents a unique opportunity for this analysis, as the survey was conducted between June 2019 and April 2021. This time frame allows for the examination of individual-level health information for both pre-COVID and post-COVID populations, as well as a comparison between states with stricter lockdown measures and those with less stringent restrictions.

3.2 Smoke and Mirrors: Effect of California Wildfires on Crime

There has been a recent surge in literature exploring the nexus of air pollution and crime Batkeyev and DeRemer (2023); Herrnstadt et al. (2021); Burkhardt et al. (2019), however only a few examine the natural causes of air pollution, such as wildfires and their association with crime. In order to bridge this gap in the literature, I intend to utilize the California wildfire season of 2020-2021, which had burnt close to seven million acres of vegetation, and look at the effect of this event on criminal activity. In order to, implement this study design, I plan on utilizing the National Incident-Based Reporting System (NIBRS), which provides incident level information on criminal activity.

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